

CLINICAL STUDY PROTOCOL SYNOPSIS

Protocol No	Sample/2026/01
Title of Study	A Retrospective Post-market Surveillance Clinical Study to Evaluate the Results of Total Knee Arthroplasty (TKA) with Investigational Total Knee System (Sponsor).
Study Design	Retrospective study to Evaluate the Results of Total Knee Arthroplasty (TKA) with the Investigational Total Knee System (Sponsor).
Investigational Device	The Investigational Total Knee System
Sponsor	Sponsor
Indications	Indicated for use only with bone cement for patients suffering from Severe knee joint pain, loss of mobility, and disability due to rheumatoid arthritis, osteoarthritis, traumatic arthritis, polyarthritis, and Correction of functional deformities. Post-traumatic loss of knee joint contour, particularly when there is patellofemoral erosion, dysfunction, or prior patellectomy, Moderate valgus, Varus, or flexion trauma. Knee fractures are untreatable by other methods.
Objectives	Primary Objective 1. To evaluate the functional performance of the Investigational Total Knee System (Sponsor). Secondary Objective 1. To evaluate the safety of the Investigational Total Knee System (Sponsor) based on complications. 2. To evaluate the survivorship of the Investigational Total Knee System (Sponsor).
Study Population	Subjects suffering from severe knee pain and disability requiring a total knee arthroplasty and underwent knee arthroplasty with the Investigational Total Knee System.
Data Source	Medical Records
Number of Subjects	The sample size has been calculated by considering the primary endpoint of the study. At least 100 Subject's data who underwent Total Knee Arthroplasty (TKA) with the investigational device, with up to 5 years of follow-up is required, including 10% inflation to allow for attrition.
Number of sites	5 to 8 sites in India
Length of the Study	Upto 1 year: includes enrollment phase and collection of a minimum of 5-year postoperative follow-up data.
Study Visits	Retrospective pre-operative, operative, and follow-up at Postoperative 6-months, Postoperative 1-year, Postoperative 3-year Postoperative 5-year.
Inclusion Criteria	1. Male or female subject ≥ 18 years old (at the time of surgery). 2. Subject had a primary diagnosis of severe knee pain and disability that require a total knee arthroplasty. 3. Subject who has undergone total knee arthroplasty with the Investigational Total Knee System.
Exclusion Criteria	1. Subject who has less than 5 years of follow-up. 2. Non-inflammatory degenerative joint disease including osteoarthritis, traumatic arthritis, or vascular necrosis. 3. History of rheumatoid arthritis and other arthritis. 4. History of preoperative severe deformity of knee alignment.

	5. History of preoperative severe limitation of motion. 6. Habitual use of narcotic pain medications before surgery or after surgery for reasons other than knee pain. 7. History of a metabolic disorder affecting the skeletal system other than osteoarthritis or osteoporosis. 8. Subject who has more than acceptable missing data for any outcome score. 9. Subject with no preoperative or postoperative records.
Study Endpoints	Primary Endpoint: • Mean change from Baseline in functional outcome measured using Knee Society Score (KSS) at each follow-up interval out to 5 years. [Knee Society Score (KSS), a physician- and a self-administered questionnaire consisting of several items to measure the outcomes and the scores ranging from 0 to 100. The scores are Symptoms Score, Satisfaction Score, Expectation Score, Walking and Standing Score, Standard Activity Score, Advanced Activity Score, Discretionary Activity Score, and Functional Activities Score.] Secondary Endpoint: • Mean change from baseline in Western Ontario and McMaster Universities Osteoarthritis (WOMAC) Index at each follow-up interval out to 5 years. [Mean Western Ontario and McMaster Universities Osteoarthritis (WOMAC) Index The Western Ontario and McMaster Universities Osteoarthritis (WOMAC) Index, a self-administered questionnaire consisting of several items (grouped into symptoms, function, daily living, pain, etc) to measure the outcomes using the scores: Total WOMAC score (sum of all scores), WOMAC pain score (range 0 to 20), WOMAC stiffness score (range 0 to 8), and WOMAC physical function score (range of 0 to 68)] • Mean change from baseline in Knee-Specific Oxford Knee Score (OKS) at each follow-up interval out to 5 years. [Oxford Knee Assessment (OKS) is a self-administered questionnaire comprising multiple-choice questions aimed to grade subjects' ability to perform daily activities. The overall score is the sum of all items and ranges from 0 to 48.] • Investigational Total Knee System revision rate at each follow-up interval out to 5 years. [Revision rate as the cumulative incidence of revision. A revision to be any procedure that involved removal of an implant (femoral component, tibial component includes a tibial tray and tibial insert, or patellar component)] • Radiographic Analysis at each follow-up interval out to 5 years [Anterior/Posterior and lateral radiographic assessments to measure Coronal and sagittal implant alignment, radiolucencies, and ballooning osteolysis]
Safety Endpoints	• Comparison of all intraoperative, early, or late complications at each follow-up interval out to 5 years [Deep infection: Defined as clinical infection with an identified organism, cellulitis was defined as superficial erythema with no apparent joint involvement which resolved with oral antibiotics wound dehiscence was secondary gaping of part of the surgical wound requiring surgical intervention.]

	Prolonged wound leakage: Defined as leakage which resolved in the absence of signs suggesting deep infection (pain, decreased range of motion, sepsis, and organism on aspiration).] • Incidence of Adverse Events by severity, relationship to knee implant, discontinuation, outcome, Serious Adverse Events (SAE), Adverse device effect (ADE), and Serious Adverse device Events (SADE).
Statistical Methods	Statistical Analysis Plan (SAP) shall include detailed statistical aspects of the study. The statistical analysis shall be performed using SAS version 9.4 or higher. Data collected in the study will be summarized. Descriptive summaries will be the basis of study reports to participants. Summaries will routinely describe categorical data as counts and percentages, and describe continuous data in the form of means, medians, standard deviations, minima, and maxima. Summaries of variables such as implant survival, or time to event will generally be described via the Kaplan-Meier method and will be accompanied by the corresponding rates (expressed as percentages). Complication data will be summarized in the form of frequencies and percentages.
Ethical Issues	The study shall commence only after written approval is obtained from the Institutional Review Board (IRB)/Independent Ethics Committee (IEC) for the submitted protocol and other relevant documents. The study shall be performed in accordance with this protocol and relevant Standard Operating Procedures (SOPs) and in compliance with applicable regulatory requirements.
Quality Standard	This study will follow ICH standards for clinical research including ICH E6 (GCP) and ICH E3 (Study Reporting); ISO 14155 standards for the conduct of the study.

APPENDIX: STUDY SCHEDULE OF EVENTS

Event	Pre-operative	Intra-operative	Post-operative			
Evaluation	3 weeks prior to surgery	Day 0	0.5 year Follow Up	1 year Follow Up	3 years Follow Up	5 years Follow Up
Allocation of Subject ID	X					
Demographic Data	X					
Inclusion/Exclusion Criteria Review	X					
Medical/Surgical History	X					
WOMAC Index	X		X	X	X	X
Knee Society Score (KSS)	X		X	X	X	X
Surgical Details		X				
Radiographic Assessments	X		X	X	X	X
Oxford Knee Score (OKS)	X		X	X	X	X
Revision Surgical Status	X					
Implant Status			X	X	X	X
Complications Assessment		X				
Concomitant Medication Review		X	X	X	X	X
Adverse Event and Serious Adverse Event Assessment		X		X	X	X